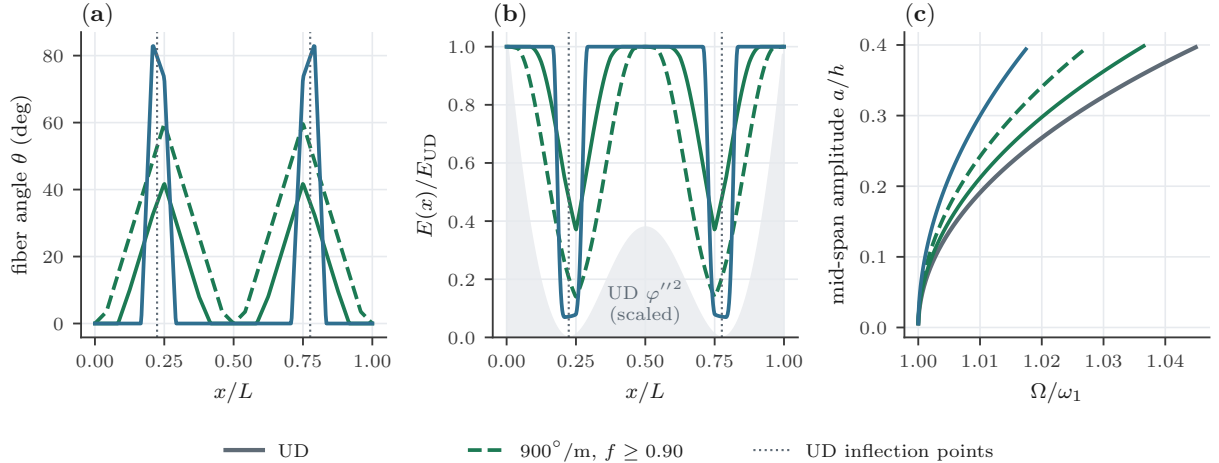

(a)
fiber angle θ (deg)
0
20
40
60
80
0.00
0.25
0.50
0.75
1.00
x/L
(b)
$E(x)/E_{\mathrm{UD}}$
0.0
0.2
0.4
0.6
0.8
1.0
UD φ''^2
(scaled)
0.00
0.25
0.50
0.75
1.00
x/L
(c)
mid-span amplitude a/h
0.0
0.1
0.2
0.3
0.4
1.00
1.01
1.02
1.03
1.04
Ω/ω_1
UD
900°/m, $f \geq 0.98$
900°/m, $f \geq 0.90$
no tow limit, $f \geq 0.98$
UD inflection points